

THADDEUS J. VANCE

Senior Director of Quality Engineering & SDET Operations

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PROFESSIONAL SUMMARY

Technically formidable and strategically focused Quality Engineering Executive with over 12 years of success in architecting end-to-end automation ecosystems for global e-commerce, fintech, and cloud infrastructure leaders. Distinguished for transforming fragmented QA departments into high-velocity SDET organizations that treat "Quality-as-Code." Expert in the deployment of AI-augmented testing frameworks, Shift-Left methodologies, and CI/CD-integrated quality gates that have consistently reduced production defect leakage by up to 85%. Thaddeus is a visionary leader who has managed annual budgets exceeding \$140M while directing cross-functional global teams of 180+ engineers across North America, EMEA, and APAC.

Proven track record in leveraging Playwright, Cypress, and custom-built Java/Python frameworks to automate testing for platforms processing over \$10B in annual revenue. A strategic advisor to C-suite stakeholders, bridging the gap between complex software architecture and business ROI. Recognized for implementing the "Vance Quality Maturity Model," which standardizes performance engineering, chaos testing, and security-integrated automation across enterprise-scale microservices. Expert in navigating SOC2, PCI-DSS, and HIPAA compliance within automated pipelines. Champion of engineering excellence who prioritizes architectural precision, horizontal scalability, and the relentless pursuit of zero-manual intervention in the release cycle.

CORE COMPETENCIES

- **Quality Strategy:** Global QE Transformation, Quality Maturity Modeling, Risk-Based Testing, Shift-Left/Shift-Right Adoption.
- **Automation Architecture:** Hybrid Framework Design, Headless Execution, Parallelized Testing, Autonomous Self-Healing Tests.
- **Leadership & P&L:** Global Team Scaling (0 to 180+), Budgetary Stewardship (\$140M+), Vendor Management, M&A Due Diligence.
- **SDET Operations:** Career Pathing for SDETs, Coding Standards for QA, Peer Review Processes, Internal Tooling Development.
- **CI/CD & DevOps Integration:** Jenkins/GitHub Actions/GitLab CI, Infrastructure-as-Code (IaC) for Test Labs, Containerized Testing.
- **Performance Engineering:** Scalability Testing, Latency Analysis, Load/Stress Modeling, Chaos Engineering, Synthetic Monitoring.
- **Specialized Domains:** API-First Testing, Microservices Quality, Mobile App Automation (iOS/Android), Security Regression Testing.
- **Data Analytics:** Predictive Defect Analytics, MTTR Optimization, Root Cause Analysis (RCA) Automation, Test Data Management.

TECHNICAL SKILLS

Automation Frameworks & UI Testing

- **Web UI:** Playwright (Expert), Cypress.io, Selenium Grid 4.0, Puppeteer, WebdriverIO, TestCafe.
- **Mobile Architecture:** Appium (Java/Python), Espresso, XCUITest, Detox, Flutter Driver.
- **Visual Testing:** Applifools Eyes (Visual AI), Percy.io, Galen Framework.
- **Cross-Device Infrastructure:** BrowserStack, SauceLabs, AWS Device Farm, LambdaTest.

Backend, API & Service Testing

- **API Testing:** RestAssured, Supertest, Postman/Newman, Karatelabs, SoapUI.
- **Contract Testing:** Pact.io, Spring Cloud Contract.
- **Messaging & Event Testing:** Kafka-Tester, RabbitMQ verification, AWS SQS/SNS Mocking.
- **Database Testing:** SQL Server, PostgreSQL, MongoDB, Cassandra (validation scripts and data seeding).

Performance, Reliability & SRE

- **Load Generation:** k6 (Go-based), Apache JMeter, Gatling (Scala), Locust.io (Python).
- **Chaos Testing:** Gremlin, Chaos Mesh, LitmusChaos.
- **Observability:** New Relic, Datadog, Prometheus, Grafana, ELK Stack (Kibana).
- **Network Analysis:** Wireshark, Charles Proxy, Fiddler.

Engineering & DevOps Stack

- **Languages:** Java (Advanced), JavaScript/TypeScript (Expert), Python (Expert), Go, SQL, Bash.
- **Containerization:** Docker, Kubernetes (K8s), Helm, Docker Compose.
- **CI/CD Pipelines:** GitHub Actions, Jenkins (Shared Libraries/DSL), GitLab CI, CircleCI, Azure DevOps.
- **Cloud Platforms:** AWS (Architect level), GCP, Azure Cloud Services.

PROFESSIONAL EXPERIENCE

Senior Director of Quality Engineering | Titan Commerce Global | 2020 – Present

Seattle, WA / Global (Multi-Billion Dollar E-commerce Marketplace | Annual Revenue: \$14B)

Strategic Mandate & Executive Leadership: Recruited to lead the 200-person Quality Engineering organization across 5 global engineering hubs. Accountable for the quality of the customer-facing storefront, payments engine, and warehouse management systems. Manage a consolidated annual budget of \$145M, including tool procurement, global headcount, and cloud execution costs.

Key Measurable Achievements:

- **Transformation to SDET Culture:** Executed a 24-month roadmap to transition the organization from a 60% manual QA model to a 95% SDET-driven automation model. Hired and trained 80+ engineers in advanced TypeScript/Playwright architecture.

- **CI/CD Pipeline Acceleration:** Integrated "Quality Gates" into the deployment pipeline that execute 5,000+ tests in under 12 minutes through massive Kubernetes parallelization. Resulted in a 400% increase in production deployment frequency.
- **Zero-Defect Black Friday:** Engineered a high-frequency load testing strategy using k6 that simulated 5 million concurrent users. Identified and mitigated 12 critical bottleneck services, resulting in \$0 in revenue loss due to downtime during peak season.
- **AI-Powered Self-Healing Tests:** Deployed an AI-driven selector engine that automatically updates test locators when UI changes are detected. Reduced test maintenance effort by 70% and eliminated "flaky test" noise by 85%.
- **Mobile Quality Overhaul:** Re-architected the Appium/Java framework for the Titan iOS/Android apps. Achieved 90% code reuse across platforms and reduced the mobile release cycle from 3 weeks to 4 days.
- **Cost Optimization:** Consolidated 12 disparate testing tools into a unified Enterprise suite, negotiating vendor contracts that saved the company \$3.5M in annual OpEx.
- **Quality Maturity Index (QMI):** Developed a proprietary QMI dashboard for the Board of Directors, providing real-time visibility into defect leakage, test ROI, and organizational risk.

Director of Quality Assurance & Automation | BlueVest Fintech | 2016 – 2020

New York, NY (High-Growth Digital Banking & Wealth Management)

Operational Leadership: Directed the QA and Automation strategy for a fast-scaling Fintech firm. Built the automation department from the ground up, scaling the team from 5 to 65 SDETs. Focused on high-reliability testing for financial transaction engines.

Key Measurable Achievements:

- **Financial Compliance Automation:** Developed an automated verification suite that validated all transaction data against SEC and FINRA regulatory requirements during the build process, ensuring 100% audit readiness.
- **Shift-Left Security:** Partnered with InfoSec to integrate DAST (Dynamic Application Security Testing) tools into the automation framework. Identified 120+ security vulnerabilities in the development phase over 4 years.
- **Database Consistency Engine:** Built a custom Python tool to verify data integrity across 400 microservices during large-scale migrations to PostgreSQL, ensuring zero data loss for 50M+ customer records.
- **Framework Reliability:** Re-engineered a legacy Selenium suite into a robust Java/TestNG framework using the Screenplay Pattern. Improved test stability from 65% to 99.7%.
- **Talent Development:** Established the "BlueVest SDET Academy," an internal program that upskilled 15 manual testers into proficient automation engineers within 6 months.
- **M&A Integration:** Led the technical quality audit for the \$800M acquisition of *WealthFlow*, successfully integrating their testing pipelines into the core BlueVest ecosystem within 90 days.

Senior Manager of Quality Engineering | CloudStack Systems | 2013 – 2016

San Francisco, CA (Enterprise Cloud Management SaaS)

Role Overview: Primary architect of the automation strategy for the CloudStack platform. Led a team of 20 SDETs in building a specialized framework for testing complex cloud orchestration workflows.

Key Measurable Achievements:

- **Scalable Test Lab:** Built a private cloud-based test lab using Docker and OpenStack, allowing for the execution of 10,000+ tests simultaneously. Reduced regression time from 48 hours to 2 hours.
- **API Contract Testing:** Introduced Pact.io for contract testing between microservices, reducing "integration" bugs in staging by 60%.
- **Shadow Deployment Testing:** Developed a "Shadow execution" engine that ran automated tests against production traffic in a non-blocking mode to identify environment-specific edge cases.
- **Customer-Centric Quality:** Integrated customer support ticket data with the QA dashboard to prioritize automation for features with the highest user-reported issues.

Automation Lead / SDET | Vertex Cybersecurity | 2011 – 2013

Austin, TX (Network Security Software)

Key Measurable Achievements:

- **Framework Creation:** Designed and implemented the first automated regression suite for the "Vertex Sentinel" product using Python and Selenium. Achieved 75% coverage in 12 months.
- **Linux Hardening Tests:** Built a library of automated security hardening scripts that verified OS-level configurations against CIS benchmarks during the appliance build process.
- **Jenkins CI Implementation:** Architected the company's first CI pipeline, automating the build-test-deploy cycle for a team of 50 developers.

KEY PROJECTS PORTFOLIO

Project Titan: The "Zero-Latency" Quality Fabric

Role: Executive Lead & Chief Architect | **Duration:** 18 Months

Objective: Eliminate the testing bottleneck for a global e-commerce engine processing 50,000 requests per second.

- **Strategy:** Deployed a distributed testing mesh on AWS EKS. Utilized gRPC for ultra-fast communication between test workers and the application under test.
- **Action:** Developed a "Predictive Execution" model that uses machine learning to analyze code changes and only run the relevant 10% of the test suite.
- **Outcome:** Achieved a 90% reduction in CI/CD execution time. Prevented an estimated \$15M in developer productivity loss per year.

Project Phoenix: Legacy Migration & Reliability Overhaul

Role: Modernization Lead | **Investment:** \$12M | **Duration:** 14 Months

Objective: Re-build a failing 10-year-old automation suite for a Tier-1 Fintech platform.

- **Strategy:** Migrated 8,000 legacy Selenium/Ruby tests to a modern TypeScript/Playwright architecture.
- **Action:** Developed a custom transpiler to automate 65% of the code conversion. Implemented strict Linting and SonarQube quality gates for test code.
- **Outcome:** Saved \$4M in annual technical debt and infrastructure maintenance. Test reliability improved from 50% to 99.9%, restoring engineering trust in automation results.

Project Aegis: Automated Security & Compliance Testing

Role: Technical Strategy Lead | **Duration:** 12 Months

Objective: Automate the verification of HIPAA and SOC2 controls across 200 microservices.

- **Strategy:** Integrated OWASP ZAP and custom security scanning scripts into the Cypress automation suite.
- **Action:** Built an "Evidence Collector" that automatically generates PDF compliance reports for auditors based on successful test executions.
- **Outcome:** Reduced manual audit preparation time by 95% and ensured continuous compliance, preventing potential regulatory fines of up to \$5M.

Operation Global Nexus: Test Data Management (TDM) Automation

Role: Lead Architect | **Outcome:** Eliminated 100% of data-related test failures.

- **Strategy:** Built an ephemeral data-seeding engine using Snowflake and Docker.
- **Action:** Created a system where every test suite "provisions" its own unique, anonymized data set and "destroys" it after execution.

LEADERSHIP & TEAM MANAGEMENT

- **Scaling High-Performance Teams:** Successfully grew the Quality Engineering organization at Titan Commerce from 15 to 180 professionals in 3 years, utilizing a "Follow-the-Sun" model with hubs in Seattle, London, and Bangalore.
- **SDET Career Framework:** Authored the "Global QE Competency Matrix," defining 6 levels of growth from Associate SDET to Principal Quality Architect. This framework became the standard for the entire engineering organization.
- **Engineering Culture:** Established "Quality Guilds" across the organization to share best practices in automation, performance, and accessibility. Resulted in a 30% increase in cross-team knowledge sharing.
- **Diversity & Inclusion in Tech:** Spearheaded a "Women in SDET" recruitment initiative, increasing female representation in the technical QA organization by 45% over 3 years.
- **Executive Stakeholder Management:** Provide bi-weekly "Quality & Risk" briefings to the CEO and Board members, aligning engineering outputs with commercial objectives.

PROCESS IMPROVEMENTS & AUTOMATION

- **Automated Root Cause Analysis (ARCA):** Built a Python-based tool that uses NLP to analyze test failure logs and automatically categorize bugs (Environment, Code, or Test). Reduced triage time from 4 hours to 5 minutes.
- **Self-Healing UI Tests:** Integrated a machine-learning locator engine into the core framework. Tests automatically "re-learn" element paths when IDs or XPath paths change, reducing maintenance tickets by 85%.
- **Chaos Injection in QA:** Introduced automated chaos injection (latency, pod failure) into the staging pipeline. Increased system resiliency, resulting in a 50% reduction in production S1/S2 incidents.
- **Shadow Deployment Pipeline:** Developed a "Shadow execution" engine that runs automated tests against production traffic in a non-blocking mode to identify environment-specific edge cases.
- **Automated Release Notes:** Integrated Jira with Jenkins to automatically generate technical release notes and quality coverage reports to stakeholders upon successful deployment.

STRATEGIC QE VISION: THE FUTURE OF QUALITY

The Transition to Autonomous Quality

I believe the next decade of quality engineering will be defined by the transition from "Automated" testing to "Autonomous" quality. We are currently building systems that don't just execute pre-written scripts, but use AI to explore the application, generate their own test cases based on user behavior data, and self-correct when code changes. My mission is to build the infrastructure that allows this "Self-Governing Quality" to exist within a high-speed CI/CD environment.

Quality as a Brand Differentiator

In a saturated SaaS market, stability and performance are the ultimate brand differentiators. I champion a philosophy where "Quality is Everyone's Responsibility." This means providing developers with the tools to own their quality and providing executives with the data to make risk-informed decisions. I move organizations away from "QA as a Filter" to "QE as a Catalyst" for speed and reliability.

Ethical AI in Testing

As we integrate Generative AI into our testing pipelines, I am focused on the ethical use of data. We must ensure that our automated systems do not introduce bias into the testing of AI-driven products and that user privacy remains the paramount concern during test data generation and RUM analysis.

BOARD ENGAGEMENT & INDUSTRY ADVISING

- **Audit & Risk Committee Reporting:** Provide quarterly briefings to the Board of Directors at Titan Commerce on cyber-resilience, system stability metrics, and the ROI of the QE investment.
- **Industry Advisor:** Currently serve as an advisory board member for three early-stage QA-Tech startups, providing feedback on AI-driven test maintenance and cloud-execution scaling.
- **Technical Due Diligence:** Acted as the primary QE evaluator for 5 successful acquisitions (totaling \$1.4B in deal value), assessing the target company's technical debt and automation maturity.

EDUCATION

Master of Science in Software Engineering & Reliability

Carnegie Mellon University, Pittsburgh, PA

- Research Focus: Reliability Modeling for Distributed Systems, Statistical Methods in QA.
- Thesis: "Scalability Limits of Cloud-Native Automation Pipelines."
- Honors: CMU Graduate Excellence Award.

Bachelor of Science in Computer Science & Mathematics

The University of Washington, Seattle, WA

- Honors: Summa Cum Laude; President of the UW Computer Science Honor Society.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Professional**
- **CKA:** Certified Kubernetes Administrator
- **ISTQB-CTEL:** Test Automation Strategy Specialist
- **Lean Six Sigma Black Belt** (Focus on Software Process Optimization)
- **CISSP:** Certified Information Systems Security Professional
- **Google Cloud Professional Cloud Architect**

TOOLS & TECHNOLOGIES REFERENCE

Automation: Playwright, Cypress, Appium, Selenium 4, XCUITest, Espresso, Detox, Pact.io.

Performance: k6, JMeter, Gatling, Gremlin, New Relic, Datadog, Grafana.

DevOps: Jenkins, GitHub Actions, GitLab CI, Docker, Kubernetes, AWS, GCP, Terraform.

Languages: JavaScript, TypeScript, Java, Python, Go, SQL, Bash.

AI-Driven QA: Appliflow, Mabl, Testim, GitHub Copilot.

Test Management: Xray (Jira), TestRail, Zephyr, Qase.io.

PUBLICATIONS & CONFERENCES

Selected Speaking Engagements

- **Keynote Speaker:** "The Death of Manual QA: Transitioning to SDET Culture," *StarWest 2023*.
- **Featured Speaker:** "Scaling Playwright to 20k Tests: Lessons from the Trenches," *TestBash UK 2022*.
- **Panelist:** "AI & The Future of Autonomous Testing," *Quality Sense Conf 2021*.
- **Workshop Lead:** "Chaos Engineering for QE Teams," *SeleniumConf Online 2020*.
- **Guest Lecturer:** "Advanced Test Architecture," *Stanford University - School of Engineering*.

Selected Peer-Reviewed Publications

- **"Predictive Defect Modeling in Microservices Architectures"**
IEEE Journal of Software Quality, 2023.
- **"Beyond Selenium: Why the Future of Testing is Headless and Native"**
The SDET Quarterly, 2022.
- **"The Cost of Flakiness: Quantifying the Impact of Unstable Tests on Revenue"**
Journal of Quality Engineering, 2021.
- **"Shift-Left Security: Automating Vulnerability Detection in the QA Pipeline"**
Cybersecurity Digest, 2020.

LANGUAGES & GLOBAL EXPOSURE

- **English:** Native Proficiency.
- **Spanish:** Professional Working Proficiency.
- **Technical Contexts:** Expert in navigating international technical cultures and labor laws for engineering recruitment in the EU, APAC, and North America.
- **Global Footprint:** Directly managed distributed teams in: USA, Canada, Poland, Germany, UK, India, and Singapore.